What is a data breach?

01 / RISK | Personal data, trade secrets or technical information reaching unauthorized parties

Main routes of leakage

External attacks: phishing, unauthorized access, exploitation of vulnerabilities and data theft accompanying ransomware attacks.

Human factors inside the organization

Misdirected email, lost devices, unauthorized removal of information, excessive access rights and inadequate management.

Cloud and generative AI use

Incorrect public-access settings in the cloud, poorly managed sharing links and entering confidential information without checking service terms or internal rules.

Impact on business: direct losses

Investigation and recovery costs, operational shutdowns, customer support, compensation for damages and more.

Medium- and long-term losses

Reduced trust among customers and business partners, lost contracts and weaker competitiveness from leaked technology.

Data breaches result from everyday operating mistakes as well as external attacks.

Key systems to prevent leakage

02 / DEFENSE | Defense in depth combines authentication, communications, endpoints, data and monitoring

DLP | Data loss prevention

Identifies confidential information and detects or restricts email transmission, copying to external media and cloud uploads according to settings.

EDR / XDR | Threat detection and response

EDR monitors endpoint behavior. XDR analyzes information across endpoints, email, identities and other sources, supporting investigation and containment.

IAM / MFA | Identity, access rights and multifactor authentication

IAM manages identities and access permissions. MFA uses multiple authentication factors to reduce unauthorized logins after a password leak.

SASE / ZTNA | Protecting communications and access

SASE integrates networking and security functions. ZTNA verifies users and devices and controls connections to required resources.

SIEM / SOC | Log analysis and monitoring operations

SIEM aggregates and analyzes logs. SOC is an organization or function responsible for monitoring, analysis and response; their roles differ.

The scope and effectiveness of each function depend on the product, settings and operating structure.

Major companies and focus areas

03 / COMPANIES | Representative related companies; listing order indicates neither ranking nor market share

Japanese companies

Digital Arts: web and email filtering, data-leakage prevention.

Trend Micro: endpoint and server protection, threat detection and response.

FFRI Security: threat analysis, Japanese-developed endpoint defense products.

Soliton Systems: authentication, access control and endpoint security.

HENNGE: authentication and access management for cloud services.

Overseas companies

Palo Alto Networks: network protection, SASE and data protection.

CrowdStrike: endpoint defense, threat detection and response, data protection.

Zscaler: zero-trust connectivity and cloud-based data protection.

Microsoft: identity management through Entra, threat protection through Defender, data protection and governance through Purview.

Operations that protect information

04 / OPERATIONS | Combine products with day-to-day management and incident response

Identify assets and information

Inventory devices, systems and storage locations; classify information by confidentiality. Clarify who can access which information.

Maintain basic safeguards

Multifactor authentication, least-privilege access, applying updates, checking email and sharing settings, and continuous employee education.

Rules for generative AI

Define which information may be entered, which services are permitted and the conditions for data retention and training use. Understand the scope of monitoring and control.

Detection and initial response

Monitor suspicious actions and communications, and identify contacts and responsible staff. Investigate the impact of incidents, contain damage and preserve evidence.

Recovery and prevention of recurrence

Verify backups and restoration procedures to prepare for business recovery. Analyze incident causes and review permissions, settings and operating rules.

Backups prepare for recovery; they do not prevent the information leak itself.